**The historical CH density instrument**

*State defaults to the historical CH density index, using the derivative instrument, historical reporting-area densities, one-step GMM and 80% known-population area coverage.*

**1. Describe historical settlement within the selected region.** Let $g$ be an MSA or a historical State-analysis group. The latter uses the dataset's county FE assignments for the territory year. Let $P_j$ and $a_j$ denote population and mapped area in square kilometres of historical reporting unit $j$ in the instrument year. Joint reports keep their combined areas. Let $s_{jg}$ be the area share allocated to member counties of $g$. Define $\ell_j = \log(P_j/a_j)$ and $w_{jg} = s_{jg}P_j / \sum_k s_{kg}P_k$. The historical CH factor is

$$H_g(\theta_0) = \sum_j w_{jg} \exp\big((\theta_0 - 1)\ell_j\big).$$

Only known positive populations enter these weights. Geographic splits change the assigned population, while preserving the original source density.

**2. Derivative instrument.** The first method uses the slope of $\log H_g$ at $\theta_0 = 1$:

$$z_g = \left.\frac{\partial \log H_g(\theta_0)}{\partial \theta_0}\right|_{\theta_0=1} = \sum_j w_{jg}\ell_j.$$

This describes log density around an average historical resident. Although $H_g(1) = 1$ everywhere, its slope differs across regions. No preliminary elasticity estimate is needed. This instrument stays fixed while GMM estimates the modern productivity elasticity $\rho = \theta - 1$.

**3. NLS-based instrument.** First estimate the modern CH equation by nonlinear least squares (NLS), using the same outcome, regional sample, controls and county adjustments as the final model. Call this estimate $\widehat{\theta}_{\mathrm{NLS}}$. Construct

$$z_g = \frac{\log H_g(\widehat{\theta}_{\mathrm{NLS}})}{\widehat{\theta}_{\mathrm{NLS}} - 1}.$$

The denominator only rescales the instrument; at $\widehat{\theta}_{\mathrm{NLS}} = 1$, use the derivative limit. GMM holds the historical instrument fixed while estimating structural $\theta$ in the full modern CH model. With one instrument and one elasticity, one-step and two-step GMM give the same solution. Inference conditions on the NLS estimate. Since NLS uses productivity, the usual fixed-instrument Stock–Wright test is not reported.

**4. Choose the coverage requirement.** Choose *80%*, *95%*, or *No coverage filter*. Divide the area with a recorded population, including recorded zeros, by the full mapped area of the member counties. Apply the selected minimum in both historical years. The State county-matching and area-weighted instruments use this same screen. CH requires positive assigned population and a finite index at every setting. For a predicted construction, coverage counts the known area of counties with the required prediction inputs. Both CH methods use the regional membership fixed by the territory year.